Structural Aliasing Analysis in Time-Series Forecasting Using Chronos

Matteo Boniotti, Gianluca Brignoli and Federico Sabbadini*

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Executive Summary

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Time-Series Forecasting

Time-series forecasting stands as a cornerstone of modern predictive analytics, vital for interpreting and anticipating the dynamic, state-dependent behaviors inherent to complex systems. Whether predicting fluctuating loads in renewable energy grids, identifying macro-economic trends, or monitoring structural health in industrial machinery, the ability to project historical sequential data into future horizons shifts organizational strategy from reactive mitigation to proactive optimization. As systems grow increasingly interconnected and generate data at unprecedented scales, the demand for forecasting frameworks capable of parsing intricate temporal dynamics has driven a massive evolution in modeling methodologies.

During the preparation of this work, the authors used AI-based tools, including GPT-5.5, Gemini 3.6 Flash, Sonnet 5, and Opus 4.6, to support language refinement and improve clarity. All generated or suggested content was carefully reviewed, revised where necessary, and validated by the authors, who assume full responsibility for the final content of this work.

Signal Representation and Aliasing Vulnerabilities

Regardless of the downstream modeling architecture utilized, the predictive fidelity of any forecasting pipeline remains strictly bounded by the digital representation of the underlying signal. This representation relies heavily on initial data acquisition, signal conditioning, and pre-processing protocols.

The fundamental mathematical boundary governing the digitization of continuous physical processes is the Nyquist-Shannon sampling theorem. The theorem dictates that a continuous-time signal containing frequency components can be perfectly captured and reconstructed without loss of information if and only if it is sampled uniformly at a sampling frequency f_s that is strictly greater than twice the highest frequency component ($f_{\max}$) present within the signal:

$$f_s > 2f_{\max}. \tag{1}$$

The threshold $f_N = \frac{f_s}{2}$ is defined as the Nyquist frequency. When a system violates this criterion, either due to an insufficient sampling rate or the absence of a proper low-pass anti-aliasing filter, high-frequency components beyond the Nyquist limit fold back into the lower spectrum. This distortive phenomenon, known as *aliasing*, introduces spurious, artificial low-frequency components that permanently mask and corrupt the true underlying dynamics of the signal.

This work addresses a distinct, architecturally induced phenomenon: *structural aliasing*. Unlike classical aliasing, structural aliasing does not arise from an insufficient sampling rate — the signal is correctly Nyquist-sampled at f_s. Instead, it emerges from the interaction between signal periodicity and the patch tokenization grid, making certain frequencies effectively invisible to the model despite being fully representable in the discrete sample sequence.

Evolution of Forecasting Methodologies

Historically, time-series analysis relied on classical statistical models such as AutoRegressive Integrated Moving Average (ARIMA) and Exponential Smoothing (ETS). These traditional methods are deeply rooted in mathematical theory, offering high interpretability, exact statistical guarantees, and robust performance on stable, low-dimensional datasets. However, they rely heavily on strict underlying assumptions such as linearity, and stationarity.

To capture complex, non-linear temporal behaviors without rigid feature engineering, the paradigm shifted toward deep learning. Early architectures leveraged Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, and Convolutional Neural Networks (CNNs) to map temporal sequences sequentially. These architectures operate on the raw scalar sequence and therefore do not introduce patch-based structural aliasing, a risk specific to the windowed tokenization strategies adopted by transformer-based foundation models.

More recently, the state of the art has been dominated by transformer-based architectures. By replacing recurrent bottlenecks with self-attention mechanisms, transformers can process sequences in parallel and explicitly compute dependencies between distant time steps. This allows the model to learn both localized high-frequency fluctuations and long-range macro-patterns simultaneously, forming highly adaptive architectures that can be fine-tuned to specific downstream industrial or scientific applications.

Representing the cutting edge of this evolution is the emergence of time-series foundation models, most notably exemplified by the *Chronos* family of frameworks. Unifying time-series forecasting with the architectures powering modern natural language processing, Chronos treats time-series values not as continuous regressions, but as discrete language tokens. The framework tokenizes real-valued historical data by quantizing values into discrete buckets via a scaling and binning operator.

This paper targets Chronos-Bolt, the Chronos 1.0 successor variant, which introduces patch-based tokenization as a key efficiency improvement in terms of sequence length and scalability over the original architecture. All formal claims in this document apply to Chronos-Bolt's patch preprocessing stage, not to the value-quantization scheme of original Chronos.

Patch-Based Tokenization: a Formal Description

Let $\mathbf{x} = \{x[n]\}_{n\in\mathbb{Z}}$ be a discrete-time signal representing the input time series. We define the patch-based tokenization operator $\mathcal{T}_{P,S}$ with patch length $P \in \mathbb{N}^+$ and stride $S \in \mathbb{N}^+$ as a mapping that transforms the signal $\mathbf{x}$ into a sequence of vector-valued tokens $\mathbf{T} = \{\mathbf{t}_k\}_{k\in\mathbb{Z}}$, where each token $\mathbf{t}_k \in \mathbb{R}^P$.

The k-th token is defined as

$$\mathbf{t}_k = \mathcal{T}_{P,S}(\mathbf{x})[k] = \begin{bmatrix} x[kS] \\ x[kS+1] \\ \vdots \\ x[kS+P-1] \end{bmatrix}. \tag{2}$$

Using element-wise indexing notation,

$$\mathbf{t}_k[m] = x[kS+m], \qquad m \in \{0,\dots,P-1\}. \tag{3}$$

Linearity of $\mathcal{T}_{P,S}$. The operator $\mathcal{T}_{P,S}$ is *linear*: since each component $\mathbf{t}_k[m] = x[kS+m]$ is a linear function of x, it follows directly that

$$\mathcal{T}_{P,S}(\alpha\, x) = \alpha\, \mathcal{T}_{P,S}(x), \tag{4}$$
$$\mathcal{T}_{P,S}(x+y) = \mathcal{T}_{P,S}(x) + \mathcal{T}_{P,S}(y). \tag{5}$$

Linearity implies superposition: the token representation of a mixture of sinusoids equals the mixture of their individual token representations. This is the theoretical foundation for the TSMixup experiments: frequency-specific aliasing effects can be isolated and recombined linearly. In matrix form, collecting K patches yields $\mathbf{P} = \mathbf{W}\cdot\mathbf{x}$ where $\mathbf{W} \in \{0,1\}^{KP\times N}$ is a binary selection matrix.

Phase-Locking and Patch Indistinguishability

Consider a continuous-time periodic signal sampled uniformly at sampling frequency f_s, yielding the discrete-time sequence $x[n]$. We isolate a fundamental harmonic component and model it as

$$x[n] = \sin\left(2\pi \frac{f}{f_s} n + \phi\right), \tag{6}$$

where f denotes the signal frequency satisfying the Nyquist criterion $f < \frac{f_s}{2}$, and $\phi \in [0, 2\pi)$ is an arbitrary phase offset.

Phase-locking occurs when the structural phase profile of the signal repeats exactly under a translational shift equal to the stride S. Evaluating a stride translation yields

$$x[n+S] = \sin\left(2\pi \frac{f}{f_s}(n+S) + \phi\right) = \sin\left(2\pi \frac{f}{f_s} n + 2\pi \frac{fS}{f_s} + \phi\right). \tag{7}$$

For exact alignment between the signal and the tokenization operator, the phase accumulation over one stride must correspond to an integer number of 2π–periods

$$2\pi \frac{fS}{f_s} = 2\pi c, \qquad c \in \mathbb{N}^+. \tag{8}$$

As a result, the signal exhibits exact stride-period translational invariance

$$x[n+S] = \sin\left(2\pi \frac{f}{f_s} n + 2\pi c + \phi\right) = \sin\left(2\pi \frac{f}{f_s} n + \phi\right) = x[n]. \tag{9}$$

Solving Equation (8) for f yields the phase-locking frequency class $\mathcal{F}_{\text{lock}}$:

$$\mathcal{F}_{\text{lock}} = \left\{ f \in \mathbb{R}^+ \;\middle|\; f = c\frac{f_s}{S},\ c \in \mathbb{N}^+,\ f < \frac{f_s}{2} \right\}. \tag{10}$$

Complete Degeneracy Classification (Three Regimes) Equation (10) captures only the *integer* case ($S \cdot f/f_s \in \mathbb{Z}^+$, full degeneracy). A complete characterisation requires analysis of rational and irrational values of the ratio $\rho = S \cdot f/f_s$:

Full degeneracy ($\rho \in \mathbb{Z}^+$): $x[n+S] = x[n]$ exactly. Consecutive tokens are identical: $\mathbf{t}_{k+1} = \mathbf{t}_k$ for all k.

Partial degeneracy ($\rho = p/q \in \mathbb{Q} \setminus \mathbb{Z}$, in lowest terms): The signal is periodic with respect to the patch grid with period q strides: $\mathbf{t}_{k+q} = \mathbf{t}_k$ for all k. Every q-th patch is identical; within a window of L tokens.

No degeneracy ($\rho \notin \mathbb{Q}$): No two patches are identical; the token sequence is quasi-periodic. Full information is preserved.

The rational case ($\rho = p/q$) is required by CW task item 2 ("analyse whether these induce degeneracies *or invariances*") and is experimentally detectable via probing the rank of $\mathbf{M}$ at frequencies $f = (p/q) \cdot f_s/S$.

The collapse of consecutive tokens is not restricted to sinusoidal signals. Consider an arbitrary discrete-time signal satisfying the translational invariance condition

$$x[n+S] = x[n], \qquad \forall n. \tag{11}$$

Taking into account two consecutive tokens, using the element-wise notation from Equation (3) and applying Equation (11) gives

$$\mathbf{t}_{k+1}[m] = x[(k+1)S + m] = x[kS + m + S] = x[kS + m] = \mathbf{t}_k[m]. \tag{12}$$

Since this equality holds for every component,

$$\mathbf{t}_{k+1} = \mathbf{t}_k. \tag{13}$$

This result holds under the *integer* phase-locking condition (full degeneracy).

Role of Patch Size P: Within-Patch Periodicity Independently of the stride-period alignment above, a single patch may exhibit *internal redundancy* when the signal period $T_0 = f_s/f$ (in samples) divides P exactly. Formally, when $P = k \cdot T_0$ for $k \in \mathbb{Z}^+$, each patch contains k complete signal cycles, so

$$\mathbf{t}_\ell[m] = \mathbf{t}_\ell[m + T_0] \quad \forall\, m \in \{0, \dots, P - T_0 - 1\}. \tag{14}$$

The patch vector is internally redundant: the model sees the waveform repeated k times within a single token and cannot distinguish it from a single-cycle patch of length T_0. Equivalently, $\mathcal{T}_{P,S}$ is non-injective on the class of T_0-periodic signals when $T_0 \mid P$.

The **combined worst-case** occurs when both conditions hold simultaneously:

$$S = m \cdot T_0 \quad \text{AND} \quad P = k \cdot T_0, \qquad m, k \in \mathbb{Z}^+. \tag{15}$$

Inter-patch degeneracy ($\mathbf{t}_{k+1} = \mathbf{t}_k$) and within-patch redundancy then co-occur, producing *maximum degeneracy*, the most severe structural aliasing regime. Stating that P is "irrelevant" to token collapse (as the preceding general argument may suggest) actively hides this second axis of degeneracy, which CW task item 2 explicitly requires characterisation of.

Characterization of the Loss of Observability

When consecutive patches become indistinguishable, a localized structural aliasing event occurs. This degradation can be characterized through three complementary perspectives.

1. **Rank Deficiency in the Embedding Space**
   Let

$$\mathbf{M} = [\mathbf{t}_k, \mathbf{t}_{k+1}, \ldots, \mathbf{t}_{k+L-1}] \in \mathbb{R}^{P \times L} \tag{16}$$

   be the local trajectory matrix.
   If two consecutive tokens become identical, then the corresponding columns become linearly dependent. Consequently, the trajectory matrix cannot achieve full rank:

$$\operatorname{rank}(\mathbf{M}) < \min(P, L). \tag{17}$$

   More precisely, the minimum achievable rank in each degeneracy regime is: full integer phase-locking $\Rightarrow \operatorname{rank}(\mathbf{M}) = 1$; rational phase-locking with denominator $q \Rightarrow \operatorname{rank}(\mathbf{M}) \leq q$; no phase-locking $\Rightarrow \operatorname{rank}(\mathbf{M}) = \min(P, L)$ (full rank). This graduated characterisation is directly testable in experiments and maps to H1/H1b.
   Repeated patch collapse systematically reduces the information content available to downstream attention layers.
2. **Content Degeneration in Self-Attention**
   If consecutive patches satisfy

$$\mathbf{t}_k = \mathbf{t}_{k+1}, \tag{18}$$

   their content representations become identical.
   (a) Content-derived attention degeneracy. For any transformer backbone, the dot-product similarity between token k and any other token j in the absence of positional modulation equals $\mathbf{q}_k^\top \mathbf{k}_j$. Under full token collapse ($\mathbf{t}_k = \mathbf{t}_{k+1}$), the content-derived query and key vectors satisfy $\mathbf{q}_k = \mathbf{q}_{k+1}$ and $\mathbf{k}_k = \mathbf{k}_{k+1}$, making rows k and $k+1$ of the content-derived attention score matrix identical. This component of the loss is valid regardless of positional encoding scheme.
   (b) RPE partial rescue. Chronos-Bolt uses a T5 backbone with relative positional biases (RPE). Under RPE, the attention bias term encodes pairwise position distance, so even identical token vectors produce different bias-corrected attention patterns — the model can still distinguish *when* a token occurred. However, RPE does not recover *what* the signal was doing: the content information lost to degeneracy is not restored by positional bias. Accurate forecasting requires both temporal position and signal content; RPE provides only the former. Framing RPE as a complete "mitigation" is therefore incorrect — it is a partial rescue that breaks the position degeneracy without addressing the content degeneracy.
3. **Loss of Sub-Stride Observability**
   The tokenization operator observes the signal only at stride-separated locations. This induces an effective observation frequency

$$f_{\text{eff}} = \frac{f_s}{S}, \tag{19}$$

   Signal dynamics occurring at temporal scales shorter than the stride interval become progressively harder to observe through the token sequence.
   The token sequence, sampled at $f_{\text{eff}} = f_s/S$, cannot represent frequencies above $f_{\text{eff}}/2 = f_s/(2S)$. This is the *Nyquist constraint on the token domain*, distinct from classical aliasing in the sample domain.

Combining with the phase-locking condition, the *blind-spot frequencies* are:

$$f_{\text{blind}} = m \cdot \frac{f_s}{S}, \qquad m = 1, 2, \ldots, \left\lfloor \frac{S}{2} \right\rfloor. \tag{20}$$

Signals at these exact frequencies are invisible to the model: their token sequences are constant or aliased, regardless of the signal's true dynamics.
Under phase-locking conditions, consecutive tokens become identical and the local representation collapses into a stationary state. Fine-grained phase evolution and high-frequency variations occurring between token boundaries therefore become hidden from the observable representation presented to the model.

Analysis of Structural Degeneracies and Invariances

When $f \in \mathcal{F}_{\text{lock}}$, the resulting behavior depends on the relationship between P and S.

Case 1: Equal Patch Size and Stride ($P = S$) When patches are contiguous and non-overlapping,

$$P = S. \tag{21}$$

Equation (11) directly implies

$$\mathbf{t}_{j+1} = \mathbf{t}_j \qquad \forall j. \tag{22}$$

The token sequence collapses into a static series of identical vectors, eliminating all observable temporal variation.

Case 2: Overlapping Windows ($S < P$) When the stride is smaller than the patch length,

$$S < P, \tag{23}$$

assume the stride-aligned invariance condition holds:

$$x[n + S] = x[n]. \tag{24}$$

Then for the k-th token,

$$\mathbf{t}_k[m] = x[kS + m], \tag{25}$$

we obtain

$$\mathbf{t}_{k+1}[m] = x[(k+1)S + m] = x[kS + m + S] = x[kS + m] = \mathbf{t}_k[m]. \tag{26}$$

Hence,

$$\mathbf{t}_{k+1} = \mathbf{t}_k. \tag{27}$$

However, this equality represents a worst-case exact phase-locking regime. In general, overlap introduces shared samples between adjacent tokens:

$$\mathbf{t}_k \cap \mathbf{t}_{k+1} \neq \emptyset, \qquad S < P, \tag{28}$$

so even under near-periodic conditions the two tokens share structure that constrains their divergence.
Therefore, overlap does not change the exact degeneracy condition, but it increases the information shared between adjacent windows, making exact collapse harder to sustain outside idealized phase alignment.

Case 3: Disjoint Windows ($S > P$) When

$$S > P, \tag{29}$$

the windows do not overlap and leave unobserved intervals of length $S - P$ between consecutive patches.

The tokenization operator only observes samples in the intervals $[kS,\, kS + P - 1]$, while the regions $[kS + P,\, (k + 1)S - 1]$ are never observed.

If the signal satisfies the stride-period condition

$$x[n + S] = x[n], \tag{30}$$

then consecutive tokens satisfy

$$\mathbf{t}_{j+1} = \mathbf{t}_j. \tag{31}$$

Independently of this condition, when $S > P$ the operator is non-injective: different signals that coincide on the observed windows but differ on the unobserved gaps produce the same token sequence.

Therefore, in this regime, information loss arises both from: (i) missing samples in the gaps, and (ii) possible periodic collapse under phase-locking.

Domain Description and Ontology Representation

The considered domain takes into account the energy generated by an asset infrastructure centered around a photovoltaic array (`pv:PhotovoltaicArray`), operating within a localized smart micro-grid ecosystem. This environment is dynamically monitored, analyzed, and managed by an autonomous AI control agent (`pv:ControlAgent`) that balances power flows and detects system failures.

Physical Assets and Grid Topology

The physical core of the system is composed of four primary entities under the physical component (`pv:PhysicalComponent`) class, which are physically connected to (`pv:connectedTo`) one another to form the micro-grid's electrical topology:

- The photovoltaic array serves as the primary generation source, defined by its physical size (`pv:panelCount`) and inverter nameplate capacity (`pv:nominalPowerW`).
- The array directly supplies power to (`pv:suppliesTo`) three downstream assets: the external national electricity grid (`pv:NationalGrid`) interface (an infinite bidirectional bus for energy exchange), local internal consumption loads (`pv:InternalLoad`) representing dynamic demand, and a local electrochemical storage unit (`pv:BatteryStorage`).
- Functionally, the battery storage unit actively buffers (`pv:buffers`) the photovoltaic array, absorbing excess generation or discharging power during deficits.

Telemetry Streams and Tokenization Layer

To evaluate system performance, data streams are continuously collected across two operational vectors:

- The photovoltaic array measures (`pv:measures`) a continuous time-series signal (`pv:Signal`). This stream encapsulates instantaneous data properties such as photovoltaic power (`pv:pvPowerW`), global horizontal irradiance (`pv:horizontalIrradianceWm2`), and tilted surface irradiance (`pv:tiltedIrradianceWm2`), alongside frequency-domain metadata including the raw sampling rate (`pv:samplingFrequencyHz`), dominant periodic cycle frequency (`pv:fundamentalFrequencyHz`), wave amplitude (`pv:amplitudeW`), and phase offset relative to epoch (`pv:phaseRad`).

- Concurrently, the array is mapped via an environmental snapshot relationship (`pv:hasWeatherObservation`) to a weather observation log (`pv:WeatherObservation`), which tracks atmospheric variables like ambient air temperature (`pv:airTemperatureC`), wind speed (`pv:windSpeedMs`), and wind direction (`pv:windDirectionDeg`).

To prepare this continuous telemetry for deep learning inference, the raw signal passes through a processing layer that tokenizes (`pv:tokenizes`) it into a patched signal representation (`pv:PatchedSignal`). This sequence chunking strategy, tailored for foundation architectures like Chronos, is explicitly governed by architectural attributes: the token block size (`pv:patchSize`), window step stride (`pv:stride`), and calculated overlap ratio (`pv:overlapRatio`).

Inferred Environmental Contexts and System Anomalies

An execution reasoning engine dynamically updates the operational metadata of the system based on physical and mathematical constraints. The system extracts a baseline generation state (`pv:GenerationState`) derived from the tilted surface irradiance feature, classifying the current solar environment into one of four mutually exclusive subclasses: full peak production clear sky (`pv:ClearSkyState`), intermittent cloud cover (`pv:PartialCloudState`), dense cloud cover overcast (`pv:OvercastState`), or zero usable irradiance nighttime (`pv:NighttimeState`).

Simultaneously, the reasoner assesses system health by validating whether the photovoltaic array exhibits an anomaly (`pv:hasAnomaly`). It detects six distinct anomaly states (`pv:AnomalyState`):

- **Clipping Anomaly (`pv:ClippingAnomaly`):** Occurs when solar power saturates at or above inverter nominal capacity during periods of high peak irradiance.
- **Inverter Trip Anomaly (`pv:InverterTripAnomaly`):** A hard breaker or component disconnection where output drops to zero despite significant solar resource availability.
- **Ramp Anomaly (`pv:RampAnomaly`):** Identified when the dynamic rate of power change (`pv:deltaPowerW`) between consecutive intervals exceeds safe structural ramp thresholds.
- **Soiling Anomaly (`pv:SoilingAnomaly`):** A persistent drop in normalized conversion efficiency, indicating panel surface contamination such as dust or soil accumulation.
- **Overheating Anomaly (`pv:OverheatingAnomaly`):** High ambient heat conditions inducing thermal derating, which depresses panel production efficiency under strong irradiance.
- **Malfunction Anomaly (`pv:MalfunctionAnomaly`):** A structural hardware fault where generation drops to a severely degraded, near-zero baseline without clear environmental causes.

Machine Learning Vulnerabilities: Aliasing Events

In parallel with physical faults, the semantic layer evaluates computational data-processing vulnerabilities. If tokenization window choices conflict with the signal's cyclical boundaries, the patched signal exhibits aliasing (`pv:exhibitsAliasing`), indicating an ongoing aliasing event (`pv:AliasingEvent`) that compromises downstream forecasting model precision:

- A frequency blind spot (`pv:FrequencyBlindSpot`) is flagged when the continuous signal component matches an integer harmonic of the patch-induced frequency (`pv:patchInducedFrequencyHz`), rendering major cyclical patterns structurally invisible to the model (Hypotheses H1 and H1b).
- A phase aliasing effect (`pv:PhaseAliasingEffect`, Hypothesis H2) occurs when an active phase offset or spatial shift (`pv:phaseOffsetRad`) misaligns with token boundaries, degrading representation fidelity.
- An overlap distortion effect (`pv:OverlapEffect`, Hypothesis H3) reveals configuration setups where a low overlap ratio indicates suboptimal stride lengths, generating context sequence block redundancy.

Closed-Loop Autonomous Action

To restore grid equilibrium and preserve physical integrity, the photovoltaic array triggers actions (`pv:triggersAction`) belonging to the agent action (`pv:AgentAction`) class hierarchy. The autonomous control agent manages the closed-loop optimization loop and performs (`pv:performs`) four specialized mitigation interventions:

- It coordinates a battery dispatch intervention (`pv:BatteryDispatchAction`) to buffer grid fluctuations under volatile weather or to substitute lost generation during unexpected inverter breaker trips.
- It deploys a generation curtailment action (`pv:CurtailGenerationAction`) to throttle active power spikes during inverter clipping or critical panel overheating states.
- It initiates demand side load adjustment changes (`pv:LoadAdjustmentAction`) to shift internal consumer consumption steps during low overcast or zero production nighttime states.
- It issues a rapid warning alert (`pv:RampAlertAction`) to stability systems the instant an active ramp rate anomaly is detected.

Structural Aliasing Expected Impact

To ground the experimental validation within the semantic framework established by the system ontology, we map the digital signal processing phenomena directly to the ontological layer. Let a continuous-time physical signal (such as `pv:pvPowerW` or `pv:tiltedIrradianceWm2`) be sampled uniformly at a rate f_s (`pv:samplingFrequencyHz`) and transformed into a tokenized sequence via a `pv:PatchedSignal` operator defined by patch size P (`pv:patchSize`) and translation stride S (`pv:stride`).

The architectural grid frequency f_p (`pv:patchInducedFrequencyHz`) governs the macro-temporal resolution of the token space and is defined as:

$$f_p = \frac{f_s}{S} \tag{32}$$

Based on this parametric environment, we define three distinct, falsifiable hypotheses to characterize the boundaries of structural aliasing.

Hypothesis 1 (H_1): Frequency-Dependent Blind Spots

- **Ontological Alignment:** `pv:FrequencyBlindSpot` (asserted via SWRL inference rules `R18` and `R19`).
- **Statement:** When the fundamental frequency f (`pv:fundamentalFrequencyHz`) of an input time-series signal matches an integer multiple of the patch-induced frequency f_p, the tokenization layer maps distinct cyclical variations into static, identical latent representations, creating discrete informational blind spots across the spectrum.
- **Mathematical Boundary:** An aliasing event is triggered when the frequency ratio satisfies the harmonic condition:

$$f = n \cdot f_p = n\left(\frac{f_s}{S}\right), \quad n \in \mathbb{N}^+ \tag{33}$$

  Under this boundary condition, the column rank of the local trajectory matrix collapses: $\mathrm{rank}(\mathbf{M}) = 1$ under full integer phase-locking, and $\mathrm{rank}(\mathbf{M}) \leq q$ under rational phase-locking with denominator q (see Section).
- **Falsifiability Criterion:** This hypothesis is falsified if empirical probing models or closed-loop reconstruction tasks demonstrate uniform representation fidelity across the operational band $[2, \frac{f_s}{2}]$ Hz, without exhibiting statistically significant localized performance degradation or info-loss anomalies at the exact integer harmonics of f_p.

Hypothesis 2 (H_2): Phase Alignment and Temporal Displacement

- **Ontological Alignment:** `pv:PhaseAliasingEffect` (asserted via SWRL inference rule `R20`).
- **Statement:** The magnitude of structural information loss at an aliased harmonic frequency is non-deterministic and varies as a function of the temporal alignment between the signal's peak-trough dynamics and the rigid boundaries of the tokenization grid.
- **Mathematical Boundary:** For a fixed harmonic frequency $f \in \mathcal{F}_{\text{lock}}$ and phase offset $\phi \in [0, 2\pi)$ in the signal
$$x[n] = \sin\left(2\pi \frac{f}{f_s} n + \phi\right), \tag{34}$$
the phase offset ϕ (radians) displaces the signal by
$$\tau = \frac{\phi\, f_s}{2\pi f} \quad \text{(samples)} \tag{35}$$
relative to the patch grid origin. Structural misalignment occurs when $\tau \bmod S \neq 0$, equivalently when
$$\phi \notin \left\{2\pi k \cdot \frac{f\, S}{f_s} : k \in \mathbb{Z}\right\}. \tag{36}$$
- **Falsifiability Criterion:** This hypothesis is falsified if information-theoretic tracking metrics (such as the Space Saving metric SV) or empirical probing classification accuracies remain strictly invariant across all randomly generated phase displacements $\phi \sim \text{Uniform}(0, 2\pi)$ at a locked harmonic frequency, proving that boundary location exercises no control over latent feature degradation.

Hypothesis 3 (H_3): Geometric Configuration and Overlap Mechanics

- **Ontological Alignment:** `pv:OverlapEffect` (asserted via SWRL inference rule `R21`).
- **Statement:** Altering the geometric relationship between the patch window length and the translational step size shifts the failure mode of the token space, moving the representation from absolute vector degeneracy to high-dimensional contextual redundancy.
- **Mathematical Boundary:** The geometric interaction is dictated by the token overlap ratio *or* (`pv:overlapRatio`):
$$or = \frac{P - S}{P} \tag{37}$$
When $or = 0$ ($P = S$), the windowing is strictly contiguous and non-overlapping; satisfying Equation (2) forces consecutive tokens to be mathematically identical ($\mathbf{t}_j = \mathbf{t}_{j+1}$), wiping out temporal gradients. When $or \to 1$ ($S \ll P$), successive patches share massive historical context; although the stride-bound phase-locking condition $x[n + S] = x[n]$ still holds, the expanded patch dimension P forces tokens to capture intra-patch phase variations that break the absolute vector identity, shifting the error into an over-smoothed informational redundancy.
- **Falsifiability Criterion:** This hypothesis is falsified if varying the architectural parameters P and S while keeping a locked signal frequency f yields an identical trajectory matrix rank profile and an identical distribution of downstream transformer attention weights, proving that the geometric layout of the patch grid does not modulate the structural aliasing expression.

Signal Generation

In order to evaluate the impact of structural aliasing on the Chronos-Bolt framework and its forecasting capabilities, we conducted a series of experiments using synthetic time-series data.

To observe the predicted values generated by Chronos-Bolt, we use three types of signal generators: a single-harmonic sinusoidal generator, a modified TSMixup generator, and a KernelSynth generator. The last two are based on methods proposed for Chronos training.[1]

Synthetic Probing Setup

Single-Harmonic Sinusoidal Generator This generator produces a simple sinusoidal signal with a single frequency component. The generated signal can be expressed mathematically as:

$$x(t) = A \cdot \sin(2\pi f t + \phi) \tag{38}$$

The continuous signal is then sampled at a specified rate to create a discrete time-series. The parameters of the generator, such as amplitude, frequency, and phase, can be adjusted to create different sinusoidal signals for testing purposes.

Light TSMixup Generator We propose a controlled probing variant of the TSMixup method described in the Chronos data augmentation procedure.[1] The original TSMixup algorithm samples subsequences from real training datasets, applies mean scaling to each sampled time series, draws mixing coefficients from a symmetric Dirichlet distribution, and returns their convex combination.

Light TSMixup preserves this stochastic mixing structure, but replaces the real dataset collection with a controlled pool of sinusoidal signals. Each base signal is defined by frequency, amplitude, phase, length, and sampling frequency. This simplification makes the generated signals easier to interpret when testing whether Chronos-Bolt exhibits frequency-dependent blind spots or phase-locking effects.

The parameters of the generator, such as the maximum number of sinusoidal components, Dirichlet concentration, target length, and sampling frequency, can be adjusted to create different synthetic signals for testing purposes. The pseudocode for the Light TSMixup generator is listed in Appendix B.

KernelSynth Generator Since TSMixup is able to produce only simple composition of sinusoidal signals, we implement KernelSynth as a more complex alternative, which can generate signals with richer frequency content and temporal dynamics. By comparing the performance of Chronos-Bolt on signals generated by both methods, we can gain insights into how structural aliasing affects the model's ability to learn and forecast time-series data with varying levels of complexity.

The KernelSynth generator is also described in the Chronos data augmentation procedure.[1] Its pseudocode and kernel bank are listed in Appendix C.

This generator produces synthetic signals by combining different kernel functions, using randomly selected weights from a normal distribution. The selected kernels are combined using sum and product operators.

As parameters, the generator takes the maximum number of kernels to combine, and the time series output length. The selected kernel functions, the pseudo-code of the method, and the parameters used are described in Appendix C.

Bayesian Analysis

Each hypothesis receives its own measurement, its own model and its own estimand. Table 2 lists the models with the value each claim predicts, Table 3 every prior, and Table 4 the rule that turns a posterior into a verdict. All are ordinary regressions; only the quantity on the left and the question the coefficient answers change between them.

Each model is built the same way: a measurement, a linear predictor μ_i that adds one offset per group the observation belongs to, and a likelihood. One coefficient is the estimand and the rest keep it unconfounded. Table 3 gives every prior with the role its parameter plays. The range over which the prior scale is varied as a sensitivity check is $\{0.25, 0.5, 1.0\}$; all are centred on no effect, so a posterior that has not moved from its prior is itself a report of an absent effect.

H1

The behavioural measurement is the forecast amplitude recovery $R = A_{\text{pred}}/A_{\text{true}}$, collected in matched triplets: each lock f_k with its two controls, all three sharing the same background realisation and the same phase, which is what makes the contrast paired. The contrast used is

$$d = \log(R_k + 0.01) - \tfrac{1}{2}\left[\log(R_- + 0.01) + \log(R_+ + 0.01)\right], \tag{39}$$

one number per triplet, negative when the lock recovers less than its neighbours. It can be formed only because the injected tone has a known frequency. Model A fits it as

$$d_i \sim \text{Student-}t_4(\mu_i,\ \sigma), \quad \mu_i = \beta_{c[i]} + u_{k[i]} + u_{b[i]}, \quad \beta_c \sim \mathcal{N}\Big(\bar{\beta} + \delta_O \widetilde{O}_c + \delta_P \widetilde{\log P}_c,\ \tau\Big). \tag{40}$$

Equation 40 has four components, each motivated separately.

- **Three offsets in μ_i:** β_c per configuration, u_k per lock harmonic and u_b per background draw, so that neither an unusual frequency nor a single corpus can be mistaken for the effect.
- **Student-t_4, not Gaussian.** Where the model rebuilds nothing, recovery is near zero and the logarithm returns very large contrasts; under a Gaussian a small number of them would determine the estimate.
- **β_c drawn around $\bar{\beta}$**, making the geometries instances of one phenomenon rather than unrelated experiments. Two consequences follow. $\bar{\beta}$ exists as a parameter, which $H1$ requires, being a claim about patch-based tokenisation and not about a single checkpoint; and each geometry contributes equal weight, so the largest design cannot dominate the estimate when the number of usable triplets varies several-fold with the in-band lock count.
- **Two configuration-level covariates**, estimable together rather than confounded because the design varies overlap and patch size independently. Overlap enters as a ratio, patch size as $\log P$ because the patch grid has spacing f_s/P and equal steps in $\log P$ are equal ratios of spacing. The tilde denotes

Run	Patch (P)	Stride (S)	overlap	tokens	$S \mid P$	$P \mid S$
1	8	4	0.50	511	yes	**no**
2	8	8	0.00	256	yes	yes
3	16	4	0.75	509	yes	**no**
4	16	8	0.50	255	yes	**no**
5	16	12	0.25	170	**no**	**no**
6	16	16	0.00	128	yes	yes
7	24	8	0.67	254	yes	**no**
8	24	12	0.50	169	yes	**no**
9	24	16	0.33	127	**no**	**no**
10	24	20	0.17	102	**no**	**no**
11	24	24	0.00	85	yes	yes
12	32	8	0.75	253	yes	**no**
13	32	12	0.62	169	**no**	**no**
14	32	16	0.50	127	yes	**no**
15	32	20	0.38	101	**no**	**no**
16	32	24	0.25	85	**no**	**no**
17	32	32	0.00	64	yes	yes

Table 1: *The extended configurations retained for the analysis. All are retrained from scratch under the same $100k$-step budget and from the same seed. $S \mid P$ means every stride lock is also a patch null, and $P \mid S$ the converse; both hold only at $P = S$, where the two branches are the same set of frequencies.*

Model	Claim	Estimand	Predicts	Input
A	$H1$ behavioural	$e^{\bar{\beta}}$: recovery at a lock / at its controls	< 1	`contrasts`
B	$H1$ representational	$e^{\theta_{lock}}$: codelength at a lock / elsewhere	> 1	`mdl_cells`
C	$H2$	σ_ϕ: spread of the deficit across phase	≈ 0	`contrasts`
D1	$H3$ location	θ_S, θ_P: dip depth on each grid, by LOO	both < 0	`collapse`
D2	$H3a$ stride branch	κ_S: measured / stride-predicted spacing	$= 1$	`sites`
D2	$H3b$ patch branch	κ_P: measured / patch-predicted spacing	$= 1$	`sites`
A′	$M1$ mitigation	δ_O: change in the deficit per unit of overlap	< 0	`contrasts`

Table 2: *The models, the quantity whose posterior decides each claim, the value the claim predicts, and the probing table each is fitted to. $H1$ is tested twice, on the forecast (A) and on the representation (B), so that a discrepancy between them stays visible. $H3$ is split into its two branches, since $\mathcal{F}_{lock}$ is a union and a frequency may belong to either. $M1$ is not a separate model but the configuration level of A, promoted to a named estimand.*

centring on the design, $\widetilde{O}_c = (O_c - \bar{O})/0.5$ and $\widetilde{\log P}_c = \log P_c - \overline{\log P}$, which leaves $\bar{\beta}$ the effect at the average configuration rather than an extrapolation to $O = 0$ and $P = 1$.

Model B poses the same question of the representation, on a frequency-local prequential codelength: at each f_c a probe separates tones at $f_c \pm 1\,\mathrm{Hz}$ from a captured internal state. Ten states are probed: the encoder [REG] token after each encoder block, the decoder state after each decoder block, the pre-projection vector and the quantile head; the decoder-side vectors are named decoder states because Chronos-Bolt carries [REG] in the encoder only. The regression is Gamma with a log link, so $e^{\theta_{lock}}$ is the factor by which the codelength of a locked frequency increases, with probe stage and geometry as zero-mean offsets since codelengths differ several-fold between stages. Deliverable 1's band tasks summarise hundreds of frequencies at once, so the lock indicator has nothing to attach to; they are kept as a cross-check.

H2

$H2$ uses the same triplets with one change. Equation 39 has already averaged over phase, so the phase index is kept instead and the response becomes the per-phase deficit $y_i = \log(R_{\text{ctrl},i} + 0.01) - \log(R_{\text{lock},i} + 0.01)$, which is d with its sign reversed and positive when the lock is worse. Model C is Model A on that response, without the configuration-level covariates and with the phase circle cut into eight slices, each given its own offset u_p:

$$y_i \sim \text{Student-}t_4(\mu_i,\ \sigma), \qquad \mu_i = \beta_{c[i]} + u_{k[i]} + u_{b[i]} + u_{p[i]}, \qquad u_p \sim \mathcal{N}(0, \sigma_\phi). \tag{41}$$

The estimand σ_ϕ is the spread of those eight offsets, and it answers the question with one number. If the deficit depends on where in its cycle the signal starts, the offsets differ and σ_ϕ is large. If the geometry alone determines it, they are equal and σ_ϕ is near zero, as $H2$ predicts. Slices are used rather than a fitted curve because they assume nothing about the form a dependence might take. Dropping u_p gives a nested null, and comparing the two by leave-one-out cross-validation (LOO) separates the size of the phase effect from the evidence that it exists.

H3

$H3$ was stated as two claims, one per branch. Both are tested on the token collapse $z_g(f)$, the dispersion of the input-patch embeddings across patches, which is zero when consecutive patches are identical. All geometries share one sweep grid: a uniform 1 Hz sweep unioned with the predicted sites of every configuration swept. No geometry is therefore scored only where its own theory predicts a dip, and non-integer sites such as $42.\overline{6}\,\mathrm{Hz}$ for $S{=}12$ are covered.

The first question is where the dips are located. Each swept frequency carries two labels, one per branch:

Model	**Parameter**	**Prior**	**Role**
A, C	$\bar{\beta}$	Student-$t_4(0, 0.5)$	population phase-lock effect, on the log-recovery scale
A	δ_O	Student-$t_4(0, 0.5)$	slope in $\widetilde{O}_c$, the centred overlap; one unit $= 0.5$ of O
A	δ_P	Student-$t_4(0, 0.5)$	slope in $\widetilde{\log P}_c$, the centred log patch size
A, C	$\tau, \sigma_k, \sigma_b, \sigma$	Half-Student-$t_4(0, 0.5)$	configuration, harmonic, background and residual scales
C	σ_ϕ	Half-$\mathcal{N}(0, 0.25)$	spread of the eight per-phase offsets
B	α_0	$\mathcal{N}(\log \bar{y}, 1)$	baseline log codelength
B	θ_{lock}	$\mathcal{N}(0, 0.5)$	log codelength expansion at a lock
B	k	Gamma$(2, 0.1)$	Gamma shape (dispersion)
B	$\sigma_{st}, \sigma_{geo}$	Half-$\mathcal{N}(0, 1)$	spread of the probe-stage and geometry offsets
D1	$\alpha_g, \theta_S, \theta_P$	$\mathcal{N}(0, 1)$	per-geometry off-grid level and the two grid labels
D1	σ	Half-$\mathcal{N}(0, 1)$	residual scale of $\log z_g(f)$
D2	κ_S, κ_P	$\mathcal{N}(0, 1)$	measured spacing per unit of predicted spacing
D2	σ_F	Half-$\mathcal{N}(0, 5)$ Hz	scatter beyond the sweep-resolution floor Δf

Table 3: *Every parameter with its prior and its role. The effect priors are those of Deliverable 1, extended by δ_P, which the design of Table 1 makes estimable.*

$$\log z_g(f) \sim \mathcal{N}(\alpha_g + \theta_S \cdot \text{OnStrideGrid} + \theta_P \cdot \text{OnPatchGrid},\ \sigma)\,. \tag{42}$$

The logarithm makes the labels multiplicative on the per-geometry off-grid level α_g. The two label coefficients θ_S and θ_P are the estimands, and $H3$ predicts both negative, so the two-label fit should win the LOO comparison against each label alone and against neither. The two labels are not competing explanations of the same dips, since a frequency may carry either or both. The two coefficients separate, however, only where exactly one label applies, which for θ_S means the geometries with $S \nmid P$.

The second question is whether the dips move. The fit above remains compatible with dips that merely sit on a fixed grid, so each branch is checked against its own scaling law. Every detected dip is assigned to the branch that predicts it, and sites belonging to both are set aside. The fundamental $\hat{f}^F_{1,g}$ of branch F in geometry g is then compared with the spacing that branch predicts, $\Delta^S_g = f_s/S_g$ or $\Delta^P_g = f_s/P_g$:

$$\hat{f}^F_{1,g} \sim \mathcal{N}\left(\kappa_F\, \Delta^F_g,\ \sqrt{\sigma_F^2 + \Delta f^2}\right). \tag{43}$$

The slope κ_F is the estimand, measured spacing per unit of predicted spacing, so $\kappa_F = 1$ denotes a branch that tracks its own arithmetic. The scatter σ_F about it is given a floor Δf, the sweep step, since a spacing read off a discrete grid is no more precise than one step; without the floor, sites landing exactly on the prediction drive σ_F to zero and the posterior degenerates.

*H*3*a*, **the stride branch** The prediction is $\kappa_S = 1$: the stride sites track f_s/S one for one, so a site that remains fixed while S changes refutes it. Identification needs a series that holds P fixed and varies S, which Table 1 supplies at P=16, 24 and 32.

*H*3*b*, **the patch branch** The prediction is $\kappa_P = 1$ in the same way. Identification needs a series that holds S fixed and varies P, which Table 1 supplies at S=8, 12 and 16.

Claim	Rule	Prior	Reads as
$H1$ supported	$\Pr(\bar{\beta} < \log 0.8 \mid D) \geq 0.95$	0.34	at least 20% attenuation at a lock
$H1$ refuted	$\Pr(\lvert\bar{\beta}\rvert < \log 1.1 \mid D) \geq 0.95$	0.14	any effect is smaller than 10%
$H2$ supported	$\Pr(\sigma_\phi < \log 1.1 \mid D) \geq 0.95$	0.30	phase moves the deficit by under 10%
$H3$ location	two-label fit wins LOO, $\theta_S, \theta_P < 0$	—	dips sit on both predicted grids
$H3a$, $H3b$	$\Pr(\lvert\kappa_F - 1\rvert < 0.1 \mid D) \geq 0.95$	0.05	that branch tracks its own parameter
$M1$ supported	$\Pr(\delta_O < 0 \mid D) \geq 0.95$, overlap term wins LOO	0.50	overlap reduces the deficit

Table 4: *Decision rules.* ***Prior*** *is the probability each rule already scored under the priors of Table 3: a posterior counts as evidence only in as much as it has moved away from that value. A claim is supported at* 0.95*, refuted at* 0.05*, and otherwise inconclusive.*

Decisions and validation

All signals are sampled at $f_s = 512\,\text{Hz}$. Pure sinusoids are swept from 2 to 250 Hz in one-hertz steps at several non-redundant starting phases, so that every tone is fully representable under the Nyquist-Shannon theorem and a loss observed at a predicted lock frequency cannot be attributed to undersampling. A single seed (42) controls all stochastic generation, ensuring full reproducibility.

The complete experimental grid is built before any data are collected: one row per combination of geometry (P, S), lock frequency, background type, and phase. Models A and C include a per-background random offset u_b with scale σ_b; this lets the posterior separate generator-specific variation from the aliasing effect itself, so that a recovery difference between Light TSMixup and KernelSynth backgrounds is attributed to the generator rather than inflating or deflating the estimate of $\bar{\beta}$. For each experimental condition, 200 background draws are archived alongside the probing results, so that the exact inputs behind every posterior can be reloaded for verification.

Every threshold in Table 4 is fixed before a posterior is inspected, and validation comes before interpretation. The analysis is implemented in a single reproducible notebook organised in five sequential parts, each corresponding to one ingredient of the Bayesian inference.

Notebook workflow

The Bayesian analysis is implemented in one reproducible notebook, `bayesian_analysis.ipynb`. Its role is not to introduce a new method after the fact, but to execute the hypotheses, measurements, priors and decision rules defined above in a fixed order. The notebook therefore has a short setup stage followed by five inferential parts: priors, observation, likelihood, posterior inference and checks. Each part writes checkpoints to disk, so that an interrupted run can resume without changing the statistical design.

Part 0 (Setup). The notebook first installs only missing dependencies, locates the repository, fixes the random seed at 42, and creates the checkpoint directories. The reportable run uses 2000 warm-up draws, 2000 posterior draws, 4 chains and `target_accept` = 0.9. The design constants are imported from `probe_lib.py`: $f_s = 512\,\text{Hz}$, a 480-sample context, a 64-sample prediction horizon, the 2–250 Hz analysis band, the retrained (P, S) geometries in Table 1, and the lock set $\mathcal{F}_{lock}$.

Part 1 (Priors). Before any Chronos output is loaded, the notebook builds the design skeleton: one planned row per geometry, lock frequency, background draw, generator and phase. It then declares the priors in Table 3, the decision thresholds in Table 4, and the sensitivity ladder $\{0.25, 0.5, 1.0\}$. Prior predictive checks are run on this skeleton rather than on a toy design, so the check asks whether the declared priors are centred on no effect, allow effects large enough to falsify or support the hypotheses, and make the decision thresholds reachable from both sides. The prior specification is saved as both JSON and Parquet, making later drift between the notebook and the report detectable.

Part 2 (Observation). This is the only GPU-dependent part, because it is the only part that loads Chronos-Bolt checkpoints. The collection module `collect.py` runs the geometries and writes five tidy tables:

`contrasts`, `mdl_cells`, `mdl_bandtasks`, `collapse` and `sites`. `contrasts` feeds Models A and C with matched lock/control recovery measurements and retains the phase index for H2. `mdl_cells` feeds Model B with frequency-local prequential codelengths. `mdl_bandtasks` is retained only as a descriptive cross-check. `collapse` feeds Model D1 with token-dispersion profiles over the union frequency grid, and `sites` feeds Model D2 with detected branch-specific spacing estimates. Collection is sharded per geometry; a completed shard is reloaded rather than recomputed, and the generated TSMixup and KernelSynth backgrounds are archived so the observations behind a posterior can be inspected.

Part 3 (Likelihood). The notebook then defines the five PyMC model factories corresponding to Models A, B, C, D1 and D2. Each factory accepts a dataframe and returns an unsampled `pm.Model`; a shared sampling wrapper applies the same NUTS settings to every fit. Hierarchical effects use non-centred parameterisations where group scales can be small. Before fitting the real observations, the notebook performs parameter recovery for Model A: it simulates contrasts on the real design matrix at known values $\bar{\beta} \in \{0, \log 0.8, \log 0.5, \log 0.1\}$, refits the model, and checks that the posterior interval covers the injected truth and is narrower than the prior.

Part 4 (Posterior inference). The fitted posteriors are either sampled or reloaded from NetCDF checkpoints. For each hypothesis, the notebook reports the estimand on its natural scale, a 95% credible interval, the posterior probability of the preregistered decision threshold, and the practical-equivalence probability. Model comparisons use leave-one-out cross-validation where the report defined competing formulations: the phase-dependent and phase-free versions of Model C, the adjusted and literal versions of Model B, the configuration-level variants of Model A for mitigation, and the competing grid labels of Model D1.

Part 5 (Checks and verdicts). The final part tests whether the posterior can be trusted before interpreting it. Convergence diagnostics require $\hat{R} < 1.01$, adequate bulk and tail effective sample size, and zero divergences. Posterior predictive checks compare data generated from Model A against the observed contrasts, both pooled and stratified by configuration and generator. Prior sensitivity refits Model A across the scale ladder and with a Gaussian likelihood replacing the Student-t_4 likelihood; the H1 conclusion is reportable only if the decision probability and credible interval remain stable. LOO tables are read using the rule $|\text{elpd\_diff}| < 2 \cdot \text{dse}$ as inconclusive. The notebook finally writes a verdict table with the three-way rule fixed before observation: supported, refuted or inconclusive.

References

[1] Abdul Fatir Ansari et al. *Chronos: Learning the Language of Time Series*. Nov. 4, 2024. DOI: `10.48550/arXiv.2403.07815`. arXiv: `2403.07815 [cs.LG]`. URL: `http://arxiv.org/abs/2403.07815` (visited on 05/13/2026). Pre-published.

Appendix A: Ontology Representation

The core ontology architecture bridges the gap between cyber-physical energy systems and deep learning signal processing domain knowledge. It categorizes the application setting into five main conceptual pillars rooted under the ontological hierarchy:

- **Physical Subsystem (`pv:PhysicalComponent`):** Represents the hardware infrastructure of the micro-grid, including the solar array, electrochemical storage, consumer loads, and the macro grid tie.
- **Cognitive Layer (`pv:ControlAgent`):** The artificial intelligence system responsible for autonomous grid optimization, load balancing, dispatch scheduling, and proactive vulnerability mitigation.
- **Signal Processing Layer (`pv:Signal`, `pv:PatchedSignal`):** Tracks the raw telemetry inputs (both time-domain features and frequency-domain components) along with their tokenized representation configurations (patch dimensions, stride values) tailored for the Chronos foundation architecture.
- **Environmental Tracking (`pv:WeatherObservation`):** Captures concurrent meteorological snapshots (air temperature, wind vectors) influencing structural physics and electrical efficiency.
- **Inferred Operational States:** Dynamic metadata classified into generation levels (`pv:GenerationState`), systemic disruptions (`pv:AnomalyState`), control directives (`pv:AgentAction`), and signal degradation hazards (`pv:AliasingEvent`).

Core Concept Taxonomy

The formal classes defined within the schema are explicitly detailed below:

`pv:PhotovoltaicArray` The primary renewable energy asset. Characterized by panel count and nameplate nominal capacity. Source of telemetry streams; associated with weather conditions and subject to environmental/electrical anomalies.

`pv:BatteryStorage` Local electrochemical reservoir capable of dynamic grid connection/isolation. Characterized by capacity and instantaneous State of Charge (SoC).

`pv:InternalLoad` Represents internal industrial or residential consumption demands. Subject to adjustable, demand-side management interventions.

`pv:NationalGrid` The external electrical macro-grid interface, acting as a bidirectional infinite bus for excess export or deficit imports.

`pv:Signal` Time-series measurement stream containing global horizontal irradiance (G_h), tilted surface irradiance (G_{tilt}), instantaneous power output (P_{pv}), and analytical frequency descriptors (sampling rate, fundamental frequency, amplitude, phase).

`pv:PatchedSignal` The Chronos-specific tokenized structure derived from a raw signal. Governed by a selected window chunking mechanism (*patchSize*, *stride*), establishing a structural *overlapRatio* and a downstream $patchInducedFrequency$.

Semantic Relations & Properties

Object Properties (Entity-to-Entity Relationships)

The physical, functional, and inferred linkages across concepts are formalized using the following object properties:

- ⟨`pv:PhotovoltaicArray`⟩ `pv:measures` ⟨`pv:Signal`⟩
- ⟨`pv:PhotovoltaicArray`⟩ `pv:hasWeatherObservation` ⟨`pv:WeatherObservation`⟩
- ⟨`pv:PhotovoltaicArray`⟩ `pv:hasGenerationState` ⟨`pv:GenerationState`⟩ *(Inferred via SWRL)*
- ⟨`pv:PhotovoltaicArray`⟩ `pv:hasAnomaly` ⟨`pv:AnomalyState`⟩ *(Inferred via SWRL)*
- ⟨`pv:PhotovoltaicArray`⟩ `pv:triggersAction` ⟨`pv:AgentAction`⟩ *(Inferred via SWRL)*

- ⟨`pv:Signal`⟩ `pv:tokenizes` ⟨`pv:PatchedSignal`⟩
- ⟨`pv:PatchedSignal`⟩ `pv:exhibitsAliasing` ⟨`pv:AliasingEvent`⟩ *(Inferred via SWRL)*
- ⟨`pv:BatteryStorage`⟩ `pv:buffers` ⟨`pv:PhotovoltaicArray`⟩
- ⟨`pv:PhysicalComponent`⟩ `pv:connectedTo` ⟨`pv:PhysicalComponent`⟩ *(Symmetric Property)*

Data Properties (Entity Attributes)

Key quantitative attributes defining system conditions include:

- **Electrical/Solar:** `pv:pvPowerW` (P_{pv}), `pv:nominalPowerW` (P_{nom}), `pv:tiltedIrradianceWm2` (G_{tilt}), `pv:deltaPowerW` (ΔP).
- **Meteorological:** `pv:airTemperatureC` (T_{air}), `pv:windSpeedMs` (W_s).
- **Tokenization Architecture:** `pv:patchSize` (N_p), `pv:stride` (S), `pv:overlapRatio` (γ), `pv:patchInducedFrequency` (f_p).
- **Signal Descriptors:** `pv:samplingFrequencyHz` (f_s), `pv:fundamentalFrequencyHz` (f_0), `pv:phaseRad` (ϕ).

Knowledge-Driven Inference System (SWRL Rules)

Real-time diagnostic, classification, and mitigation logic are automated using a rule-based engine. The standard logical form of these production rules is detailed below:

Environmental Context Extraction

$$\begin{aligned}\texttt{PhotovoltaicArray}(?pv) \wedge \texttt{measures}(?pv, ?s) \wedge \texttt{tiltedIrradianceWm2}(?s, ?g) \wedge (?g \geq 800.0) \\ \longrightarrow \texttt{hasGenerationState}(?pv, \texttt{ClearSky})\end{aligned} \tag{44}$$

$$\begin{aligned}\texttt{PhotovoltaicArray}(?pv) \wedge \texttt{measures}(?pv, ?s) \wedge \texttt{tiltedIrradianceWm2}(?s, ?g) \\ \wedge (?g \geq 150.0) \wedge (?g < 800.0) \longrightarrow \texttt{hasGenerationState}(?pv, \texttt{PartialCloud})\end{aligned} \tag{45}$$

$$\begin{aligned}\texttt{PhotovoltaicArray}(?pv) \wedge \texttt{measures}(?pv, ?s) \wedge \texttt{tiltedIrradianceWm2}(?s, ?g) \\ \wedge (?g > 0.0) \wedge (?g < 150.0) \longrightarrow \texttt{hasGenerationState}(?pv, \texttt{Overcast})\end{aligned} \tag{46}$$

$$\begin{aligned}\texttt{PhotovoltaicArray}(?pv) \wedge \texttt{measures}(?pv, ?s) \wedge \texttt{tiltedIrradianceWm2}(?s, ?g) \wedge (?g \leq 0.0) \\ \longrightarrow \texttt{hasGenerationState}(?pv, \texttt{Nighttime})\end{aligned} \tag{47}$$

Cyber-Physical Anomaly Diagnostics

- **Inverter Clipping Anomaly (R5):**

$$\begin{aligned}\texttt{PhotovoltaicArray}(?pv) \wedge \texttt{measures}(?pv, ?s) \wedge \texttt{pvPowerW}(?s, ?p) \wedge \texttt{nominalPowerW}(?pv, ?nom) \\ \wedge (?p \geq ?nom) \wedge \texttt{tiltedIrradianceWm2}(?s, ?g) \wedge (?g > 800.0) \\ \longrightarrow \texttt{hasAnomaly}(?pv, \texttt{Clipping})\end{aligned} \tag{48}$$

- **Inverter Trip Anomaly (R6):**

$$\begin{aligned}\texttt{PhotovoltaicArray}(?pv) \wedge \texttt{measures}(?pv, ?s) \wedge \texttt{pvPowerW}(?s, ?p) \wedge (?p \leq 0.0) \\ \wedge \texttt{tiltedIrradianceWm2}(?s, ?g) \wedge (?g > 200.0) \longrightarrow \texttt{hasAnomaly}(?pv, \texttt{InverterTrip})\end{aligned} \tag{49}$$

- **Ramp Event Anomaly (R7):**

$$\texttt{PhotovoltaicArray}(?pv) \wedge \texttt{measures}(?pv, ?s) \wedge \texttt{deltaPowerW}(?s, ?dp) \wedge (?dp > 500.0) \longrightarrow \texttt{hasAnomaly}(?pv, \texttt{Ramp}) \tag{50}$$

- **Soiling Performance Deficit Anomaly (R8):**

$$\begin{aligned} &\texttt{PhotovoltaicArray}(?pv) \wedge \texttt{measures}(?pv, ?s) \wedge \texttt{tiltedIrradianceWm2}(?s, ?g) \wedge (?g > 400.0) \\ &\wedge \texttt{pvPowerW}(?s, ?p) \wedge (?p > 100.0) \wedge \texttt{swrlb:divide}(?r, ?p, ?g) \wedge (?r < 0.3) \\ &\longrightarrow \texttt{hasAnomaly}(?pv, \texttt{Soiling}) \end{aligned} \tag{51}$$

- **Thermal Overheating Derating Anomaly (R9):**

$$\begin{aligned} &\texttt{PhotovoltaicArray}(?pv) \wedge \texttt{hasWeatherObservation}(?pv, ?w) \wedge \texttt{airTemperatureC}(?w, ?t) \wedge (?t > 35.0) \\ &\wedge \texttt{measures}(?pv, ?s) \wedge \texttt{tiltedIrradianceWm2}(?s, ?g) \wedge (?g > 600.0) \\ &\wedge \texttt{pvPowerW}(?s, ?p) \wedge \texttt{swrlb:divide}(?r, ?p, ?g) \wedge (?r < 0.4) \\ &\longrightarrow \texttt{hasAnomaly}(?pv, \texttt{Overheating}) \end{aligned} \tag{52}$$

Autonomous Closed-Loop Control Actuation

Based on inferred operating profiles, actions are dynamically mapped to handle grid stresses:

$$\texttt{pv:hasAnomaly}(?pv, \texttt{pv:Ramp}) \longrightarrow \texttt{pv:triggersAction}(?pv, \texttt{pv:RampAlert}) \tag{53}$$

$$\texttt{pv:hasAnomaly}(?pv, \texttt{pv:Clipping}) \vee \texttt{pv:hasAnomaly}(?pv, \texttt{pv:Overheating}) \tag{54}$$

$$\longrightarrow \texttt{pv:triggersAction}(?pv, \texttt{pv:CurtailGeneration}) \tag{55}$$

$$\texttt{pv:hasAnomaly}(?pv, \texttt{pv:InverterTrip}) \vee \texttt{pv:hasGenerationState}(?pv, \texttt{pv:PartialCloud}) \tag{56}$$

$$\longrightarrow \texttt{pv:triggersAction}(?pv, \texttt{pv:BatteryDispatch}) \tag{57}$$

$$\texttt{pv:hasGenerationState}(?pv, \texttt{pv:Overcast}) \vee \texttt{pv:hasGenerationState}(?pv, \texttt{pv:Nighttime}) \tag{58}$$

$$\longrightarrow \texttt{pv:triggersAction}(?pv, \texttt{pv:LoadAdjustment}) \tag{59}$$

Chronos Structural Aliasing Hypotheses

The key scientific contribution of this architecture lies in identifying information loss bounds during time-series patching. The model detects structural blind spots via frequency domain reasoning:

Hypothesis H1 & H1b: Patch-Induced Frequency Blind Spots

The patch tokenization process introduces an artificial operating frequency profile f_p, mathematically defined by the relationship:

$$f_p = \frac{f_s}{S} \tag{60}$$

Where f_s represents the raw Nyquist sampling rate and S denotes the token step stride.

The appropriate detection tolerance derives from the frequency resolution of the context window of length N_{context} samples:

$$\delta f \approx \frac{f_s}{N_{\text{context}}}. \tag{61}$$

For Chronos-Bolt default context lengths (512–2048 samples), this gives $\delta f \in [f_s/2048,\ f_s/512]$. The tolerance fraction $\varepsilon = \delta f / f_p = S/N_{\text{context}}$ must be validated empirically; whether $\pm 10\%$ is wider or narrower than this resolution depends on the specific S and N_{context} chosen.

- **Primary Blind Spot (H1):**

$$\begin{aligned} &\texttt{Signal}(?s) \wedge \texttt{tokenizes}(?s, ?ps) \wedge \texttt{fundamentalFrequencyHz}(?s, ?f_0) \wedge \texttt{patchInducedFrequencyHz}(?ps, ?f_p) \\ &\qquad \wedge \texttt{swrlb:divide}(?r, ?f_0, ?f_p) \wedge (?r \geq 1 - \varepsilon) \wedge (?r \leq 1 + \varepsilon) \\ &\qquad \longrightarrow \texttt{exhibitsAliasing}(?ps, \texttt{H1\_BlindSpot}) \end{aligned} \tag{62}$$

- **Secondary Harmonic Blind Spot (H1b):**

$$\begin{aligned} &\texttt{Signal}(?s) \wedge \texttt{tokenizes}(?s, ?ps) \wedge \texttt{fundamentalFrequencyHz}(?s, ?f_0) \wedge \texttt{patchInducedFrequencyHz}(?ps, ?f_p) \\ &\qquad \wedge \texttt{swrlb:divide}(?r, ?f_0, ?f_p) \wedge (?r \geq 2 - \varepsilon) \wedge (?r \leq 2 + \varepsilon) \\ &\qquad \longrightarrow \texttt{exhibitsAliasing}(?ps, \texttt{H1b\_BlindSpot}) \end{aligned} \tag{63}$$

Hypothesis H2: Phase Aliasing Effect

Vulnerabilities appear when temporal offsets misalign with chunk window anchors, corrupting early representation extraction features ($0 < \phi < S$):

$$\begin{aligned} &\texttt{Signal}(?s) \wedge \texttt{tokenizes}(?s, ?ps) \wedge \texttt{phaseRad}(?s, ?\phi) \wedge \texttt{stride}(?ps, ?S) \\ &\qquad \wedge (?\phi > 0.0) \wedge (?\phi < ?S) \longrightarrow \texttt{exhibitsAliasing}(?ps, \texttt{H2\_Phase}) \end{aligned} \tag{64}$$

Hypothesis H3: Overlap Information Distortion

When the overlap configuration ratio ($\gamma = \frac{N_p - S}{N_p}$) drops below a critical information-theoretic bound ($\gamma < 0.5$), consecutive contextual patches display heavy text-token gaps, rendering trend forecasts highly unstable:

$$\texttt{PatchedSignal}(?ps) \wedge \texttt{overlapRatio}(?ps, ?or) \wedge (?or < 0.5) \longrightarrow \texttt{exhibitsAliasing}(?ps, \texttt{H3\_Overlap}) \tag{65}$$

Appendix B: Light TSMixup Pseudocode

Algorithm 1 Light TSMixup: Controlled Sinusoidal Time Series Mixup

Require: Sinusoidal pool $\mathcal{P} = \{p_1, \ldots, p_{N_p}\}$, where each p_n is defined by frequency f_n, amplitude A_n, phase ϕ_n, source length T_n, and sampling frequency f_s; maximum components $K = 10$; symmetric Dirichlet concentration $\alpha = 1.5$; minimum and maximum augmented lengths ($l_{\min} = 128, l_{\max} = 2048$).

Ensure: An augmented time series.

$k \sim U\{1, K\}$ ▷ number of sinusoidal components to mix
$l \sim U\{l_{\min}, l_{\max}\}$ ▷ length of the augmented time series
for $i \leftarrow 1$ to k **do**
    $n \sim U\{1, N_p\}$ ▷ sample a sinusoidal source
    $x_{1:l}^{(i)} \leftarrow \text{SampleSegment}(p_n, l, f_s)$ ▷ generate or crop a length-l segment
    $\tilde{x}_{1:l}^{(i)} \leftarrow x_{1:l}^{(i)} \Big/ \left(\frac{1}{l} \sum_{j=1}^{l} \left| x_j^{(i)} \right| \right)$ ▷ apply mean scaling
end for
$[\lambda_1, \ldots, \lambda_k] \sim \text{Dir}([\alpha_1 = \alpha, \ldots, \alpha_k = \alpha])$ ▷ sample mixing weights
return $\sum_{i=1}^{k} \lambda_i \tilde{x}_{1:l}^{(i)}$ ▷ take a weighted combination of scaled signals

Appendix C: KernelSynth Pseudocode and Kernel Bank

Algorithm 2 KernelSynth: Synthetic Data Generation using Gaussian Processes

Require: Kernel bank $\mathcal{K}$ listed in Table 5; maximum kernels per time series $J = 5$; generated length $l_{\text{syn}} = 1024$; optional numerical jitter $\epsilon = 0$.

Ensure: A synthetic time series $x_{1:l_{\text{syn}}}$.

$j \sim U\{1, J\}$ ▷ sample the number of kernels

$\{\kappa_1(t,t'), \ldots, \kappa_j(t,t')\} \overset{\text{i.i.d.}}{\sim} \mathcal{K}$ ▷ sample kernels with replacement

$t \leftarrow \text{linspace}(0, 1, l_{\text{syn}})$

$\kappa^\star(t,t') \leftarrow \kappa_1(t,t')$ ▷ initialize the composite covariance

for $i \leftarrow 2$ to j **do**

  $\star \sim U\{+, \times\}$ ▷ sample a binary composition operator

  $\kappa^\star(t,t') \leftarrow \kappa^\star(t,t') \star \kappa_i(t,t')$ ▷ compose kernels

end for

if $\epsilon > 0$ **then**

  $\kappa^\star(t,t') \leftarrow \kappa^\star(t,t') + \epsilon I$

end if

$x_{1:l_{\text{syn}}} \sim \mathcal{GP}(0, \kappa^\star(t,t'))$ ▷ sample from the GP prior

return $x_{1:l_{\text{syn}}}$

Table 5: *Kernel bank $\mathcal{K}$ used by KernelSynth.*

Kernel	Formula	Hyperparameters
Constant	$\kappa_{\text{Const}}(x,x') = C$	$C = 1$
White Noise	$\kappa_{\text{White}}(x,x') = \sigma_n \cdot \mathbf{1}_{x=x'}$	$\sigma_n \in \{0.1, 1\}$
Linear	$\kappa_{\text{Lin}}(x,x') = \sigma_0^2 + x \cdot x'$	$\sigma_0 \in \{0, 1, 10\}$
RBF	$\kappa_{\text{RBF}}(x,x') = \exp\left(-\frac{\|x-x'\|^2}{2l^2}\right)$	$l \in \{0.1, 1, 10\}$
Rational Quadratic	$\kappa_{\text{RQ}}(x,x') = \left(1 + \frac{\|x-x'\|^2}{2\alpha}\right)^{-\alpha}$	$\alpha \in \{0.1, 1, 10\}$, $l = 1$
Periodic	$\kappa_{\text{Per}}(x,x') = \exp\left(-2\sin^2\left(\pi\frac{\|x-x'\|}{p/l_{\text{syn}}}\right)\right)$	$p \in \{24, 48, 96, 168, 336, 672,$ $7, 14, 30, 60, 365, 730,$ $4, 26, 52, 4, 6, 12, 4, 40, 10\}$